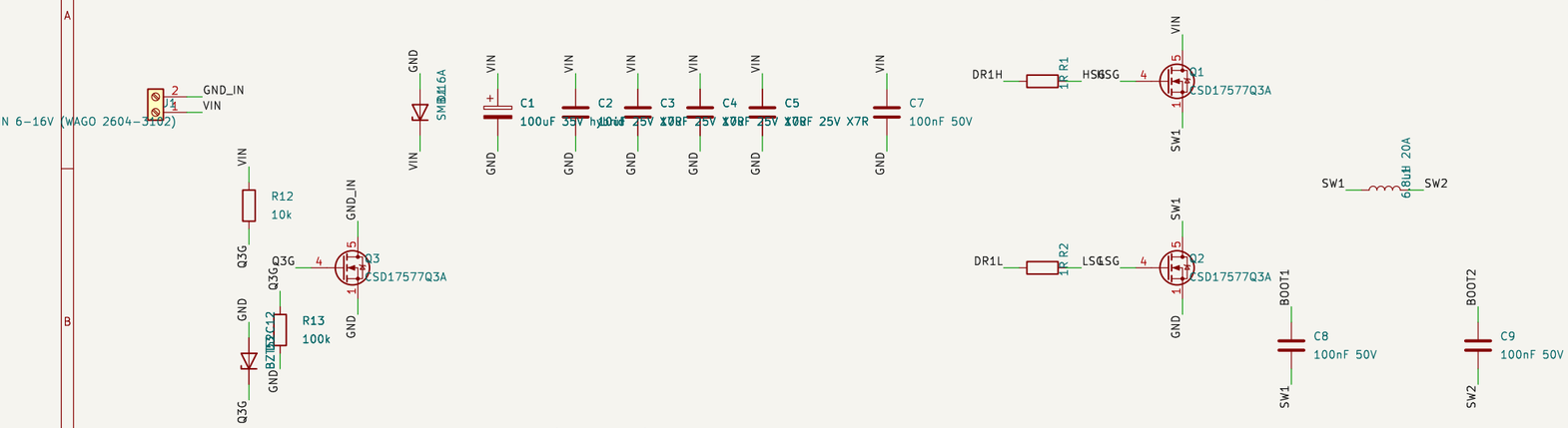
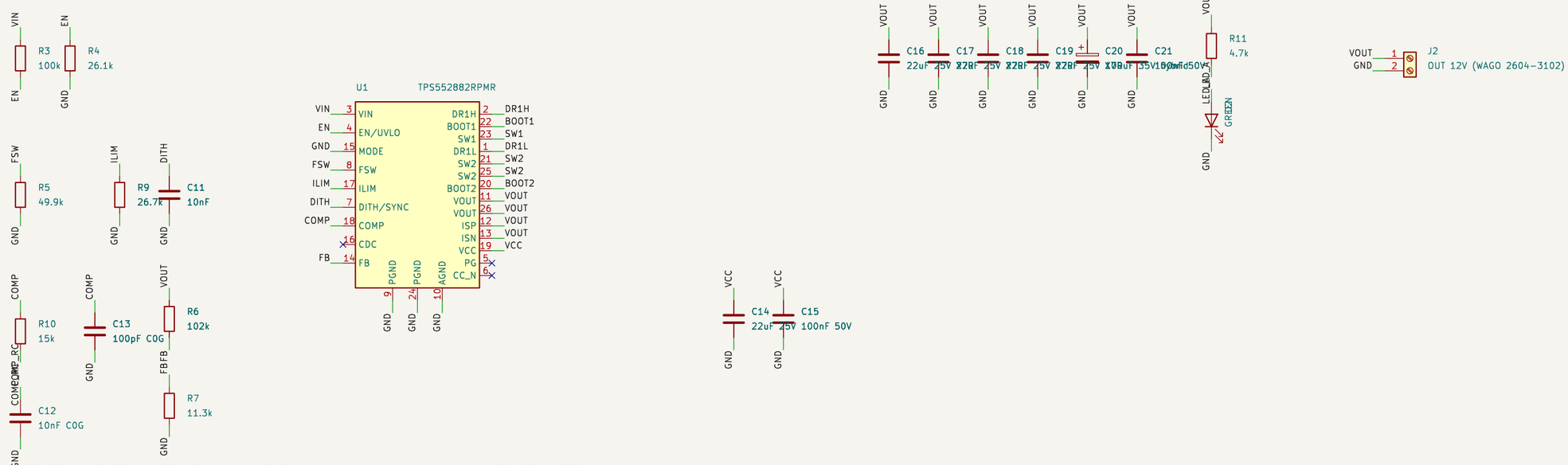
FRC 12 V PoE-switch supply — TPS552882 4-switch buck-boost, 6–16 V in -> 12.0 V / 5 A out

Input: WAGO J1 from a PDH/PDP breaker slot (10–20 A). Q3 = low-side reverse-polarity FET, D1 = 16 V TVS. Buck leg uses external FETs Q1/Q2; boost leg FETs are inside U1.



UVLO rising = 1.23 V x (1 + R3/R4) = 5.9 V FSW: R5 49.9k = 400 kHz ILIM: R9 26.7k ~ 12.4 A MODE: R8 0R = internal LDO, forced PWM Output: to Linovision POE-Switch0504GDF PWR1 (DC 9–57 V). At 12 V input the switch's PoE budget is 60 W.



VOUT = 1.2 V x (1 + R6/R7) = 12.03 V. ISP/ISN tied to VOUT (output current sense unused). PG, CC_N, CDC not used.

TPS552882 standalone, 6–16V in, 12V 5A out

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Title: FRC PoE switch 12V buck-boost

Size: A3	Date: 2026-11-10	**Rev: A**
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